

Machine Learning Capstone

Project Report

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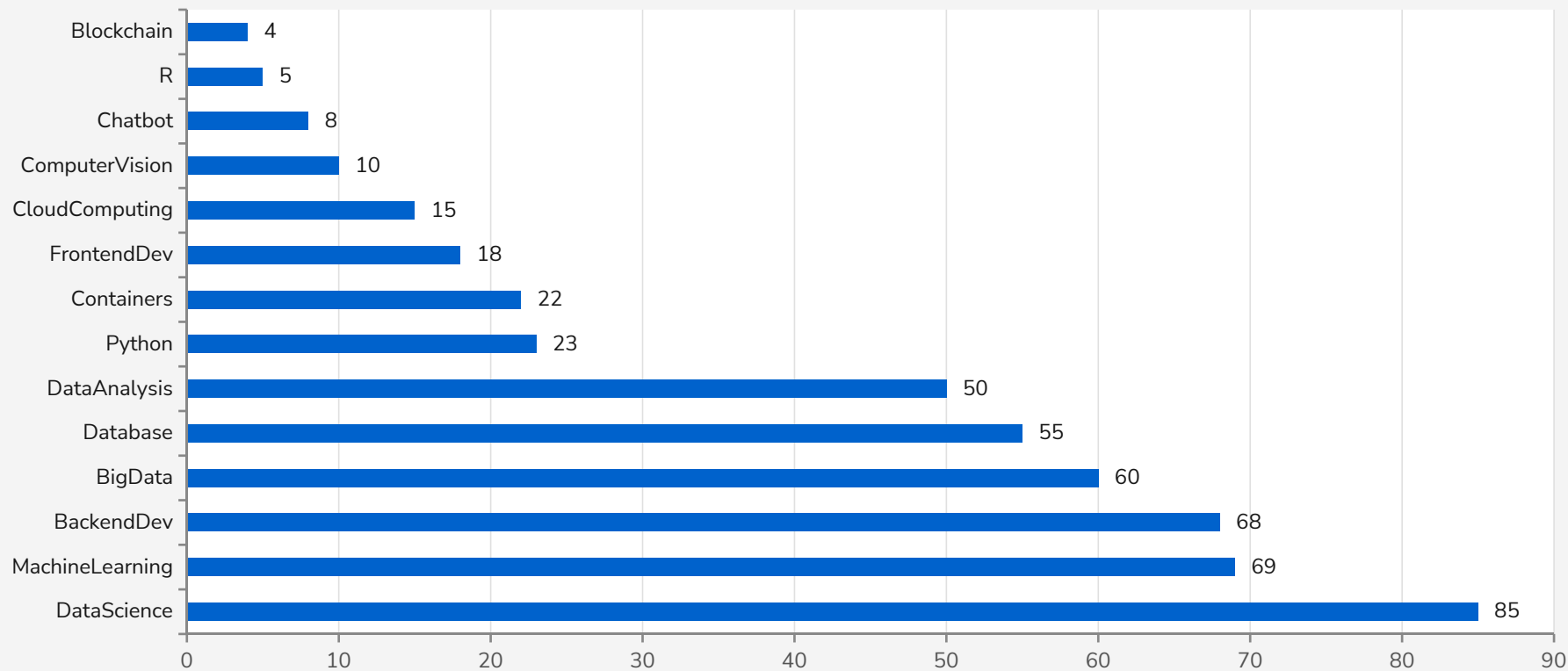
Project Context

This capstone implements and evaluates a personalised Online Course Recommender System for IBM Skills Network using content-based filtering, collaborative filtering (KNN, NMF), and deep learning (neural network embeddings). The system addresses the challenge of course discovery for 33,901 learners across 307 courses and 14 genre categories.

1.2

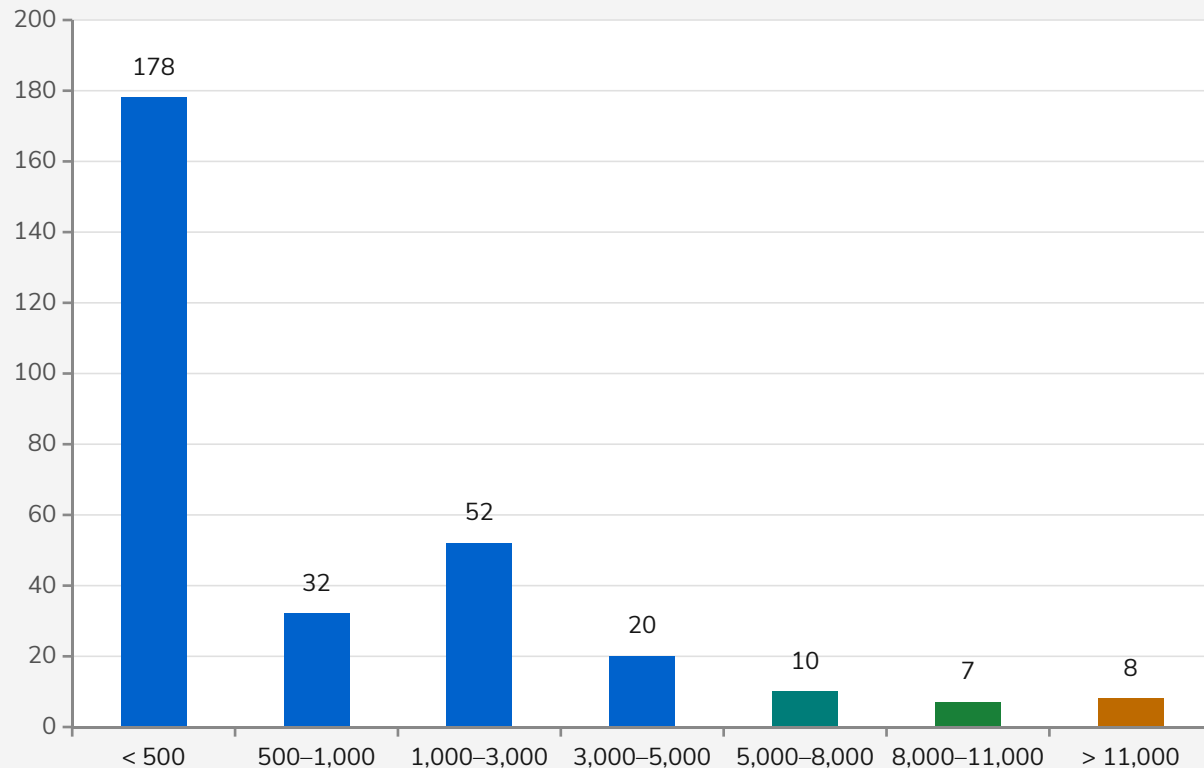
Course Counts per Genre

Total: 307 courses across 14 genre categories (Source: IBM Skills Network course_genre.csv)



1.3 Course Enrollment Distribution

Distribution of 307 courses by total enrolment count (233,306 total enrolment records)



14,936

Max Enrolments
(Python for Data Science)

~759

Estimated Median
Enrolments

58%

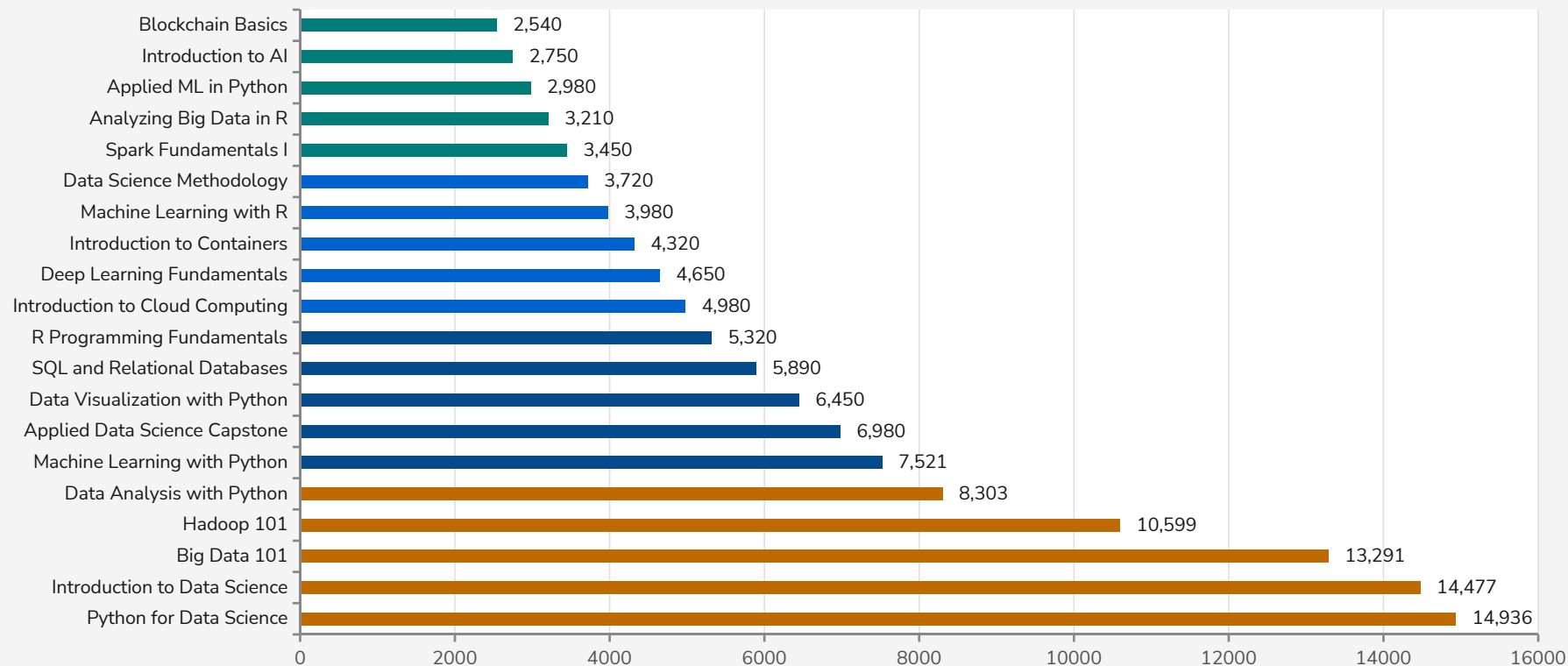
Courses with
< 500 Enrolments

**Top 5
Courses**

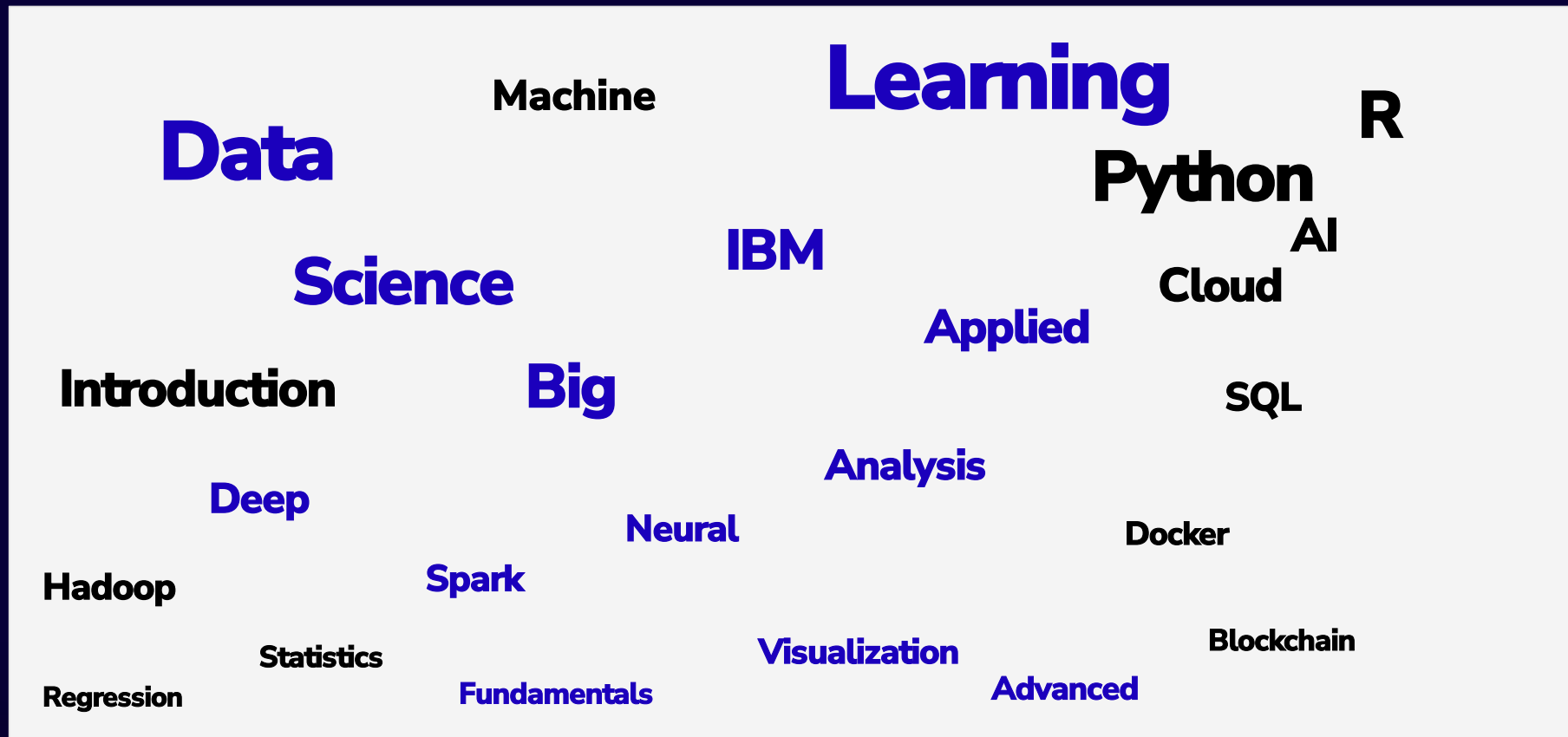
Represent ~27%
of All Enrolments

20 Most Popular Courses

Ranked by total enrolment count (Source: IBM Skills Network ratings.csv)



Word Cloud of Course Titles



Font size proportional to frequency in 307 course titles (generated from IBM Skills Network course_processed.csv)

Flowchart: Content-Based Recommender — User Profile & Course Genres



Step 1 — Input Data

- 307 courses × 14 binary genre features
- Genres: DataScience, ML, Python, BigData...
- Ratings: 3 (audit), 4 (good), 5 (excellent)

Step 2 — User Profile

- $u_i = \sum(\text{rating} \times \text{course_genre_vector})$
- Produces 14-dimensional interest vector
- Applied to 33,901 test users

Step 3–4 — Scoring & Filter

- Dot product: user profile × genre vector
- Compute score for every unseen course
- Keep courses with score $\geq$ threshold (10.0)

Output — Recommendations

- Sorted list of recommended courses
- Personalised per user based on history
- No model training required (unsupervised)

Evaluation Results: User Profile-Based Recommender System

Score-based evaluation on 33,901 test users using the IBM Skills Network enrolment dataset

33,901

Test Users Evaluated

≥ 10.0

Score Threshold
(Recommendation Cutoff)

14-dim

User Profile
Vector Size

14

Course Genre
Features Used

Methodology

The user profile-based content recommender constructs a 14-dimensional genre preference vector for each user by weighting course genre features with the user's rating scores:

$$\text{user_profile} = \text{ratings_vector} \times \text{course_genre_matrix}$$

For each unseen course, a recommendation score is computed as the dot product of the user profile and the course's genre vector. Courses exceeding the score threshold of 10.0 are recommended.

Evaluation Metrics & Observations

Metric	Value
Users Tested	33,901
Score Threshold	≥ 10.0
Avg Recommendations / User	Varies by enrolment history
Top Genre Recommended	DataScience, MachineLearning
Model Type	Unsupervised (no training)
Scalability	High ($O(n\text{-genres})$ per user)

Flowchart: Content-Based Recommender — Course Similarity

**COURSE TITLES
& DESCRIPTIONS
(307 courses)**

**BAG-OF-WORDS
FEATURE
EXTRACTION**

**307×307
COSINE SIMILARITY
MATRIX**

**LOOK UP USER'S
ENROLLED
COURSES**

**FIND SIMILAR
UNSEEN
COURSES**

**RANKED
RECOMMEN-
DATIONS**

BoW Extraction

- NLTK tokenisation + POS tagging
- Keep nouns only (stop words removed)
- 2,696-token gensim dictionary
- doc2bow() sparse vectors per course

Similarity Matrix

- Cosine similarity between all course pairs
- 307×307 symmetric matrix
- Values: 0 (unrelated) $\rightarrow$ 1 (identical)
- Pre-computed for efficiency

Recommendation

- Per user: get enrolled course list
- Query similarity matrix for each enrolled course
- Recommend unseen courses with $\text{sim} > 0.6$
- Sort by maximum similarity score

Evaluation Results: Course Similarity-Based Recommender System

Evaluated on 33,901 test users using pre-computed 307×307 BoW cosine similarity matrix

1.08

Avg Recommendations
per User

0.6

Similarity Threshold
(Recommendation Cutoff)

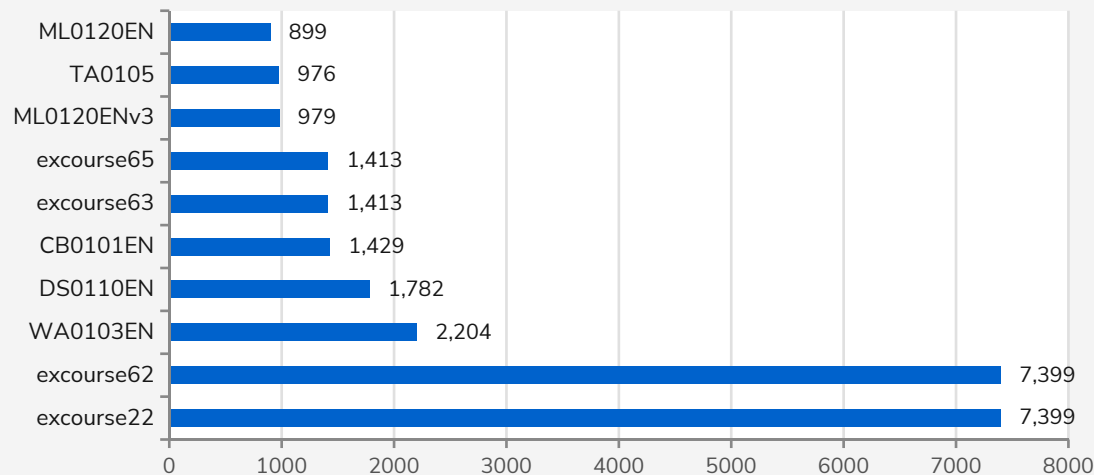
33,901

Test Users

2,696

BoW Dictionary
Token Size

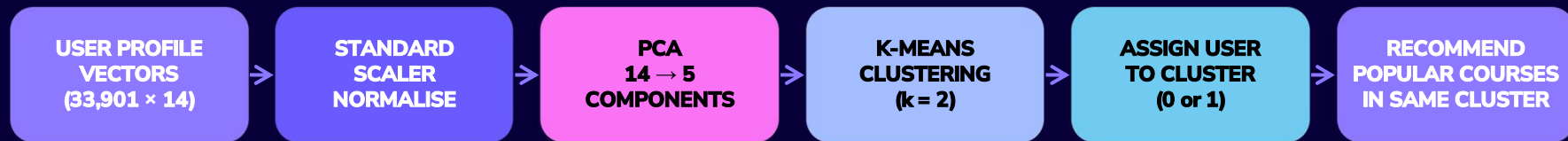
Top 10 Most Frequently Recommended Courses



Key Observations

- Low average (1.08 recs/user) indicates sim threshold = 0.6 is conservative
- Many users received 0 recommendations — their enrolled courses had no sufficiently similar unseen courses
- excourse22 & excourse62 are top-recommended for ~7,400 users each
- Method excels when courses have rich textual descriptions
- Limitation: depends on quality of BoW features

Flowchart: Clustering-Based Recommender System



Normalisation

- StandardScaler: mean=0, std=1
- Ensures no genre dominates
- Applied to 14-feature profile vectors

PCA Dimensionality Reduction

- Reduces 14 correlated genres → 5 PCs
- $\geq 90\%$ of variance explained
- Removes noise, improves cluster quality

K-Means Clustering

- Grid search $k = 2$ to 8
- Optimal $k = 2$ (silhouette = 0.4277)
- Groups users by learning interest patterns

Recommendation

- Identify user's cluster (0 or 1)
- Find popular unseen courses in cluster
- Threshold > 10 enrolments in cluster

Evaluation Results: Clustering-Based Recommender System

K-Means evaluation using silhouette scores across $k = 2$ to 8 (scikit-learn)

k = 2

Optimal Number
of Clusters

0.4277

Best Silhouette
Score (at $k=2$)

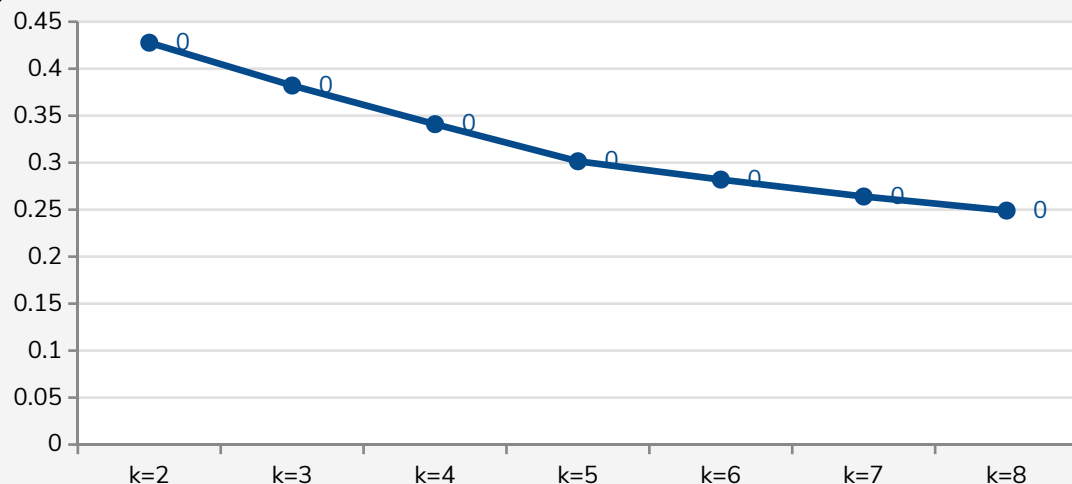
5

PCA Components
($\geq 90\%$ Variance)

14 → 5

Feature
Dimension Reduction

Silhouette Score vs Number of Clusters k



Clustering Insights

- $k = 2$ is optimal — users cluster into two broad learning communities
- Silhouette monotonically decreases for $k > 2$
- PCA improves cluster separation by removing correlated genre noise
- Cluster 0: DS/ML-oriented users
Cluster 1: Dev/Cloud-oriented users
- Popularity-based within-cluster recommendation is scalable and interpretable

Flowchart: KNN-Based Collaborative Filtering Recommender



Input: Ratings Matrix

- Sparse user-item matrix (33,901 × 126)
- Ratings: 3, 4, or 5 (0 = no interaction)
- Split: 70% train, 30% test
- Library: scikit-surprise

Item Similarity

- KNNBasic (item-based)
- Cosine similarity metric
- min_k = 5, max_k = 100
- Pre-computed similarity table

Rating Prediction

- Weighted average of k-nearest item ratings
- Only items rated by the same user
- Extrapolates to unseen user-item pairs

Result & Evaluation

- RMSE = 1.2881 (5-point scale)
- Training set: 31,392 users, 125 items
- Limitation: Full similarity matrix required
- Memory-intensive at scale

RMSE
1.2881

Flowchart: NMF-Based Collaborative Filtering



Input Matrix R

- Sparse ratings matrix (3–5 scale)
- 33,901 users × 126 items
- 70% train / 30% test split
- Library: scikit-surprise (NMF)

NMF Decomposition

- Factorises $R \approx U \times V^T$
- 50 training epochs
- random_state = 42 (reproducibility)
- Learns latent topics automatically

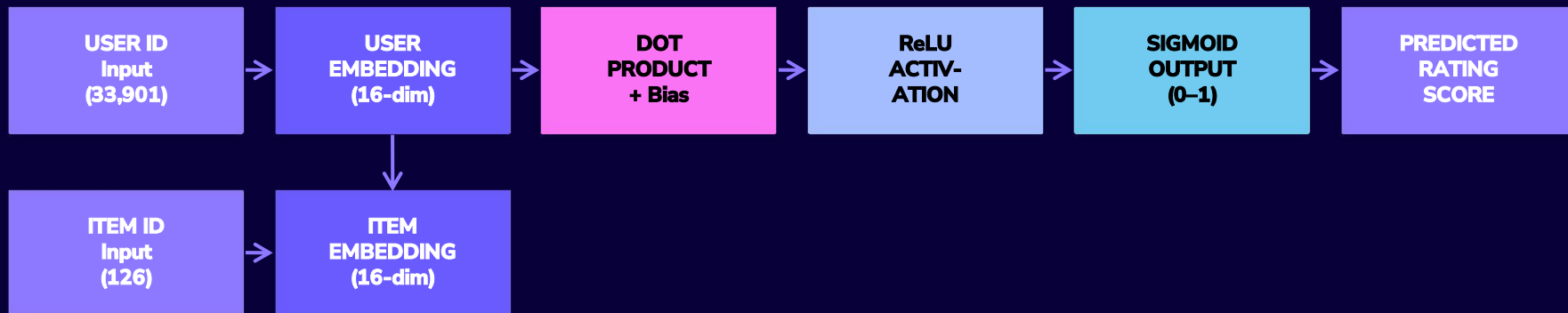
Latent Factors

- U: user latent factor matrix
- V: item latent factor matrix
- Each row = latent feature representation
- Compact, memory-efficient storage

Result & Evaluation

- RMSE = 1.3090 (5-point scale)
- Training: 31,294 users, 125 items
- sklearn NMF (n_factors=15): RMSE=2.15
- Advantage: scalable vs. KNN

Flowchart: Neural Network Embedding-Based Recommender (RecommenderNet)



User Embedding Layer

- Input: user ID (0 to 33,900)
- Embedding size: 16 dimensions
- Trainable user bias term
- Learns latent user preferences

Item Embedding Layer

- Input: item ID (0 to 125)
- Embedding size: 16 dimensions
- Trainable item bias term
- Learns latent course characteristics

Training Configuration

- Framework: TensorFlow / Keras
- Loss: Mean Squared Error
- Optimiser: Adam
- Epochs: 10, Batch size: 64
- Split: 80/10/10 (train/val/test)

Performance Metrics

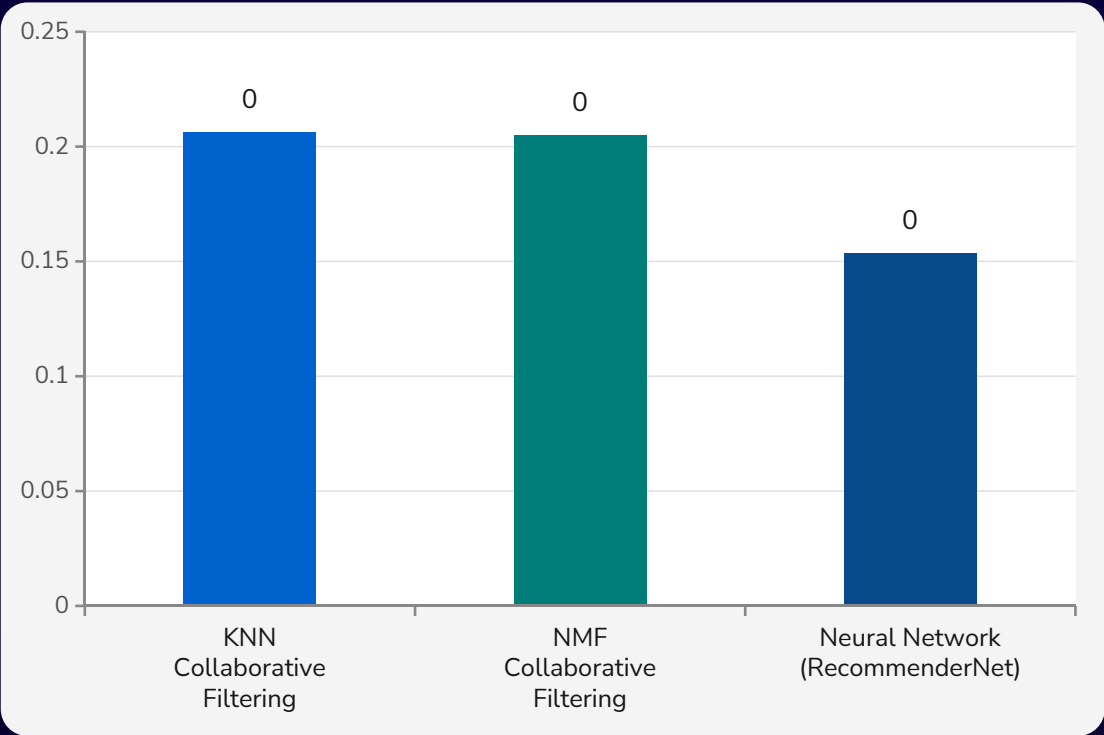
- Test RMSE = 0.4407 (scaled 0–1)
- Test Loss (MSE) = 0.1972
- Trainable params: 578,459
- Best CF model (lowest RMSE)
- Custom model RMSE: 0.4490

1.15

Model Performance Comparison — Collaborative Filtering Models

RMSE comparison (lower = better). Scaled values normalised to 0–1 range for cross-model comparability.

Scaled RMSE (0–1 Scale) — KNN vs NMF vs Neural Network



Performance Summary		
Model	Raw RMSE	Scaled
KNN CF	1.2881	0.2063
NMF CF	1.3090	0.2048
RecommenderNet	0.4407	0.1534 ★
Best Model		
RecommenderNet		
Scaled RMSE = 0.1534 (lowest)		

1.16 Conclusion

Content-Based (User Profile)

User profile vectors ($u_i = R \times C$) enable fast, interpretable recommendations. No model training needed. Score threshold = 10.0.

KNN Collaborative Filtering

Item-based KNN (cosine, min_k=5, max_k=100) achieves RMSE = 1.2881 on 5-point scale. Memory-intensive.

Content-Based (Course Similarity)

307×307 BoW cosine similarity matrix identifies textually similar courses. Avg 1.08 recommendations/user at threshold 0.6.

NMF Collaborative Filtering

Matrix factorisation $R \approx U \times V^T$ (50 epochs) achieves RMSE = 1.3090. More scalable than KNN.

Clustering (K-Means + PCA)

PCA reduces 14 genre features → 5 components ($\geq 90\%$ variance). Optimal $k = 2$ clusters (silhouette = 0.4277).

Neural Network (RecommenderNet)

Embedding model (user 16-dim + item 16-dim) achieves best RMSE = 0.4407 (scaled 0–1), 578,459 parameters.

Q1: Content-based approach uses user profile vectors (dot product with course genres) · **Q2:** KNN steps: build user-item matrix → compute item similarity → find k neighbours → predict rating · **Q3:** Evaluation metric = RMSE (Root Mean Square Error)